Volatility Models

Chapter 9

Nicolas Cachanosky

Department of Economics and Finance

Outline

- Stock returns and the limits of the mean model
- Volatility clustering and the ARCH model
- The GARCH(1,1) model
- Asymmetric volatility
- Dynamic conditional correlation
- GARCH-based Value-at-Risk

Stock returns and the limits of the mean model

Section 1

Stock returns and the limits of the mean model

- The efficient market hypothesis (EMH)
  - If a predictable pattern existed in stock prices — i.e., a higher price on Tuesdays — what would happen?
  - EMH (Fama, 1970)
    - Asset prices already reflect all available information. Any predictable pattern would be exploited by investors who buy before a predicted rise and sell before a predicted fall. In a competitive, well-informed market, no arbitrage opportunities would survive.
  - Let p_t be a stock price on period t; under the weak-form EMH:
    - $r_t = \log\left(\frac{p_t}{p_{t-1}}\right)$
    - $\mathbb{E}[r_t|\mathcal{F}_{t-1}] = \mu$
      - r_t: log-returns, μ: constant risk premium (the compensation for holding a risky asset)
    - Everything beyond μ should be unforecastable
  - An investment can regularly do as good, or worse, than he market, but it cannot do better

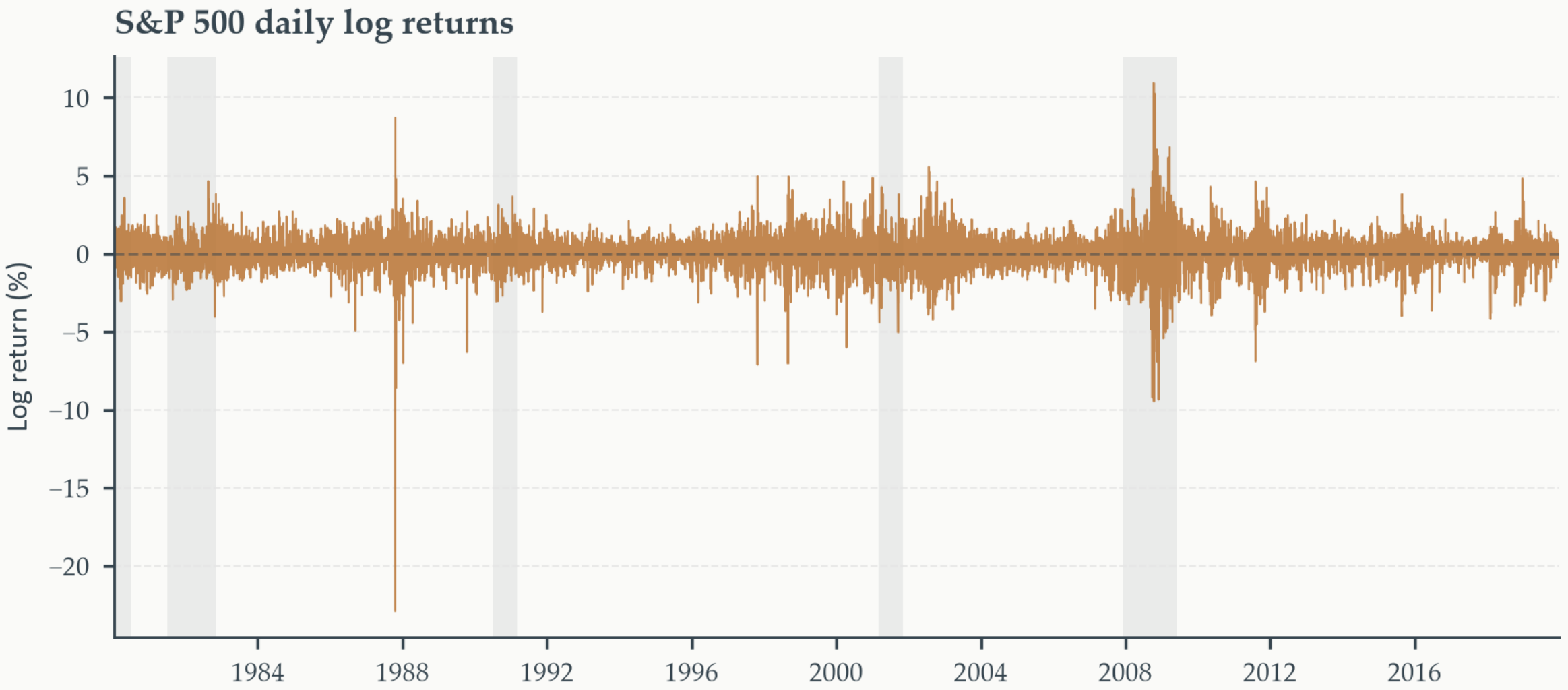
Stock returns and the limits of the mean model

- EMH versus the random walk
  - A random walk says $r_t = r_{t-1} + \varepsilon_t$
  - White noise requires ε_t to be serially uncorrelated **and** have no predictable structure of any kind — linear or nonlinear function of past returns can forecast it
  - EMH requires **white noise** innovations, not merely a random walk: There is no trend (drift) to predict, and no pattern in the noise to exploit

Stock returns and the limits of the mean model

- We worked through SARIMAX, VAR, and VECME models
- What does EMH imply for these techniques?
  - If markets are efficient, none of these models can forecast stock returns
- EMH is about the **conditional mean**, nothing more, nothing less
  - EMH → $\mathbb{E}[r_t|\mathcal{F}_{t-1}] = \mu$
  - What about $VAR(r_t|\mathcal{F}_{t-1})$?
  - EMH does not required the conditional variance to be constant
    - Knowing that around a certain event volatility will increase implies more risk and a higher rate of return, but no arbitrage opportunity
    - EMH does not rule out volatility being forecastable

Figure 9.1 — S&P 500 daily log returns, 1980–2019



Stock returns and the limits of the mean model

- We have a new "object" of study
  - Conditional heteroskedasticity
  - $\{r_t\}$ is conditionally heteroskedastic if $Var(r_t|\mathcal{F}_{t-1}) = \sigma_t^2$ is time-varying
  - Split variance into a homoscedastic component and a time-varying component
    - A series can be **unconditionally homoscedastic** while still having a time-varying conditional variance
    - Unconditional variance: $Var(r_t) = \sigma^2$ → this is your long-run average variance
    - Conditional variance: $Var(r_t|\mathcal{F}_{t-1}) = \sigma_t^2$ → this is conditional on (a) information known $(\mathcal{F})$ and (b) period t

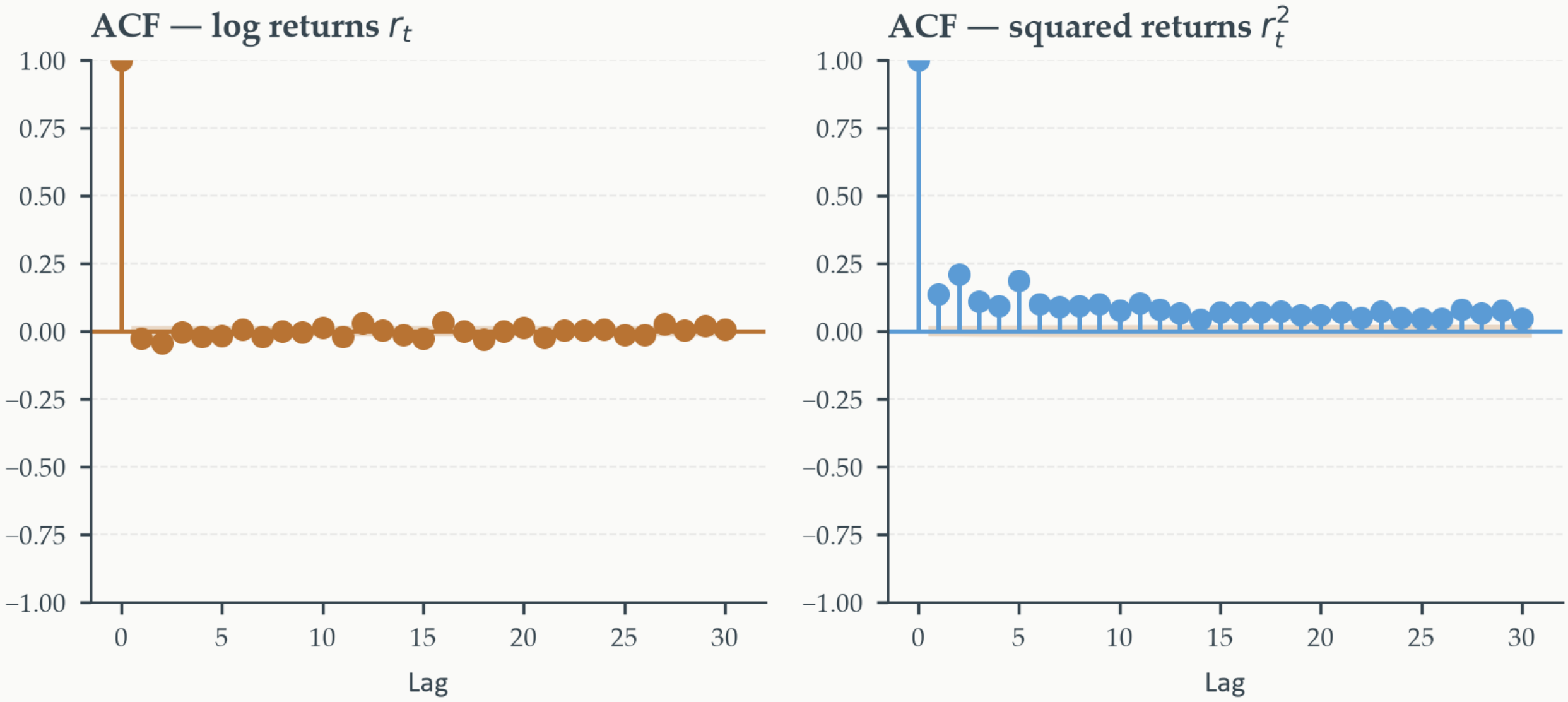
Volatility clustering and the ARCH model

Section 2

Volatility clustering and the ARCH model

- Two plots
  - ACF of r_t → no significant spikes at any lag
  - ACF or r_t^2 → slow decay; sign of an AR(p) process
- How you get a volatility model?
  - Move your regression from r_t to r_t^2
  - Now you have an **Autoregressive Conditional Heteroskedasticity (ARCH) model**

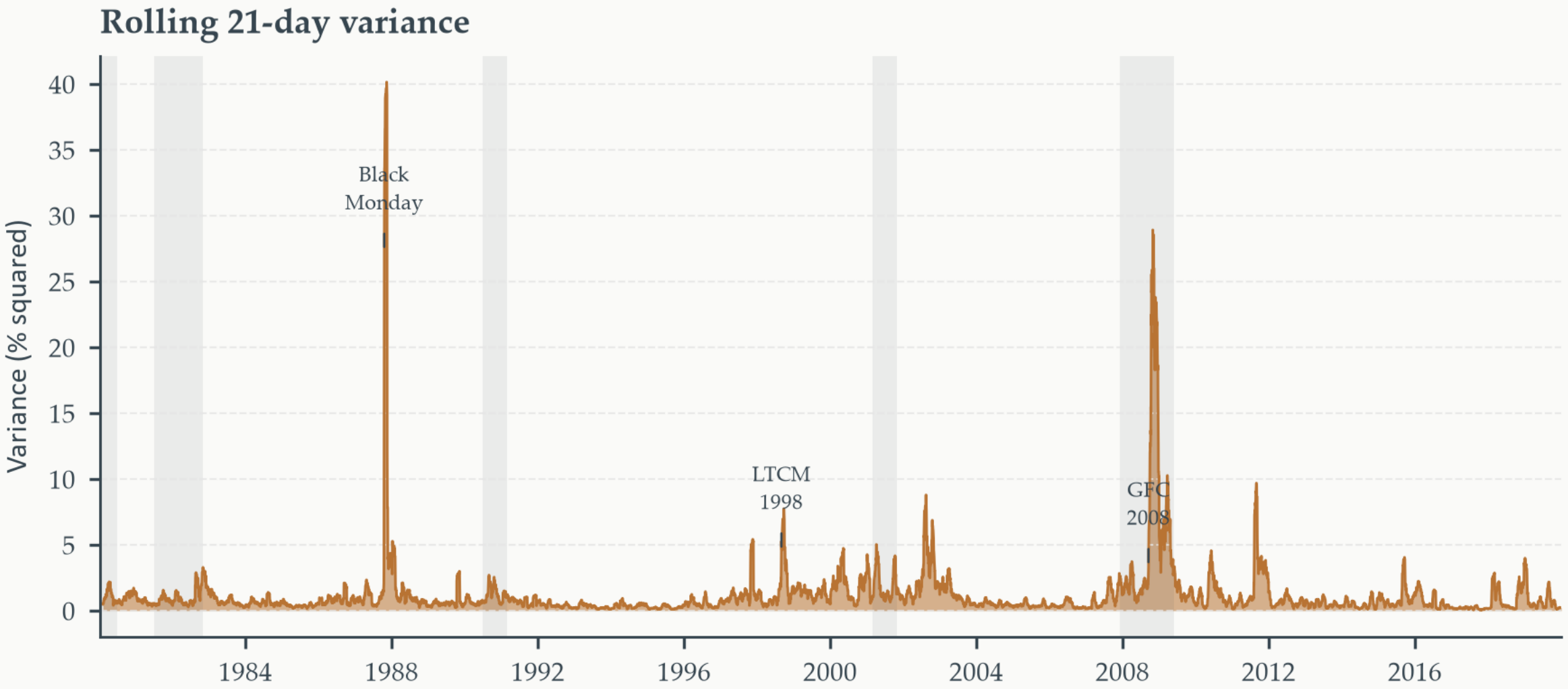
Figure 9.2 — ACF of returns vs. squared returns



Volatility clustering and the ARCH model

- The ARCH(q) model set up
  - Write the returns as mean plus innovations
    - $r_t = \mu + \varepsilon_t$
  - Decompose the innovation into a time-varying scale and standardized shock
    - $\varepsilon_t = \sigma_t z_t, \;\; z_t \sim N(0,1)$
  - The conditional variance is a weighted sum of q past squared innovations
    - $\sigma_t^2 = \omega + \left(\alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-q}^2\right) = \omega + \left(\sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2\right), \quad \omega > 0, \alpha_i \geq 0$

Figure 9.3 — Rolling variance of S&P 500 returns, 1980–2019



Volatility clustering and the ARCH model

- The ARCH-LM test
  - H_0: All q coefficients are jointly equal to zero (no ARCH effects)
  - H_a: At least one $a_q \neq 0$
  - Test statistic
    - $T \cdot R^2 \xrightarrow{d} \chi_q^2$

Volatility clustering and the ARCH model

- Example
  - Download S&P500 daily returns
  - De-mean the returns
  - Run the ARCH-LM test for $q = 10$ lags
    - $LM = 879.96$
    - $p \approx 0$
    - Reject the null of homoskedasticity → the conditional variance is time-varying

Volatility clustering and the ARCH model

- The limits of the ARCH(q) model
  - A typical ARCH model depicts a low decay on its ACF
    - r_t^2 may required 20 or more lags
  - A model may require 21+ (ω plus 20 α) coefficients, and all of them should be positive
  - The same problem than the AR(p) ↔ MA(∞) and AR(∞) ↔ MA(q) tension
  - A fix to a MA(∞) model is to fix an ARMA(1,1) model. Can we do a similar trick with the ARCH model?

The GARCH(1,1) model

Section 3

The GARCH(1,1) model

- Remember the following:
  - $\varepsilon_t = \sigma_t z_t, \quad z_t \sim N(0,1)$
  - $\sigma_t^2 = \omega + \left(\alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-q}^2\right)$
- Add a lagged conditional variance to the ARCH equation
  - Today's variance depends on **all** past squared shocks with geometrically defining weights — You need only one lagged term!
    - $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$
    - You go from 21+ terms to 3 (ω, α, β)

The GARCH(1,1) model

- Reading the GARCH(1,1) model
  - $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$
    - $\omega > 0$: the **variance floor**
    - $\alpha \geq 0$: the **shock impact** — how strongly yesterday's shock impacts today's variance
    - $\beta \geq 0$: the **variance persistence** — how much of yesterday's variance carries forward (the "AR/memory" process)

The GARCH(1,1) model

- A GARCH(1,1) is a time-series model, what about stationarity?
  - Define the centered variance $\tilde{\sigma}_t^2 = \sigma_t^2 - \sigma^2$
  - Substitute and rearrange the GARCH(1,1) model
    - $\alpha + \beta < 1 \rightarrow \sigma^2 = \frac{\omega}{1-\alpha-\beta}$
    - The stability conditions do not change

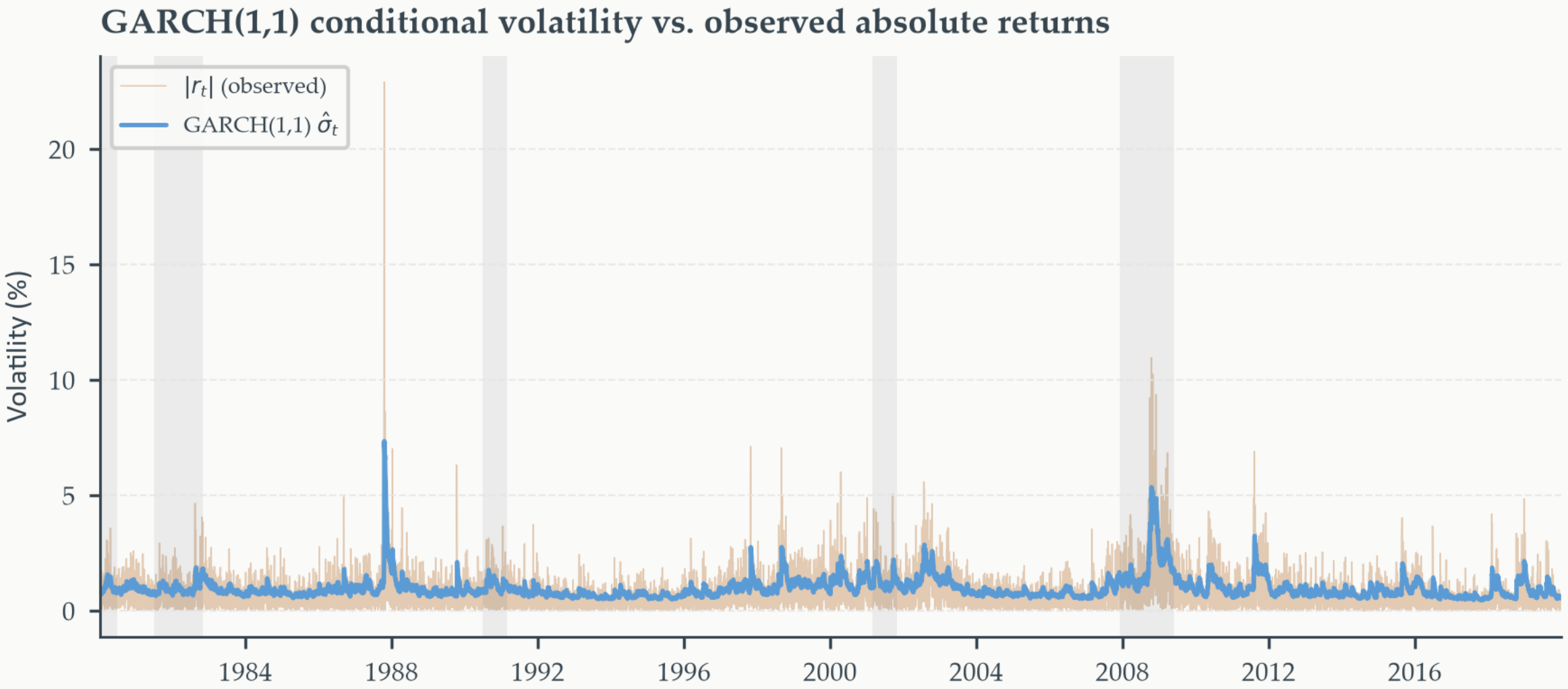
The GARCH(1,1) model

- You can also predict decay
  - The Half-Life of a variance shock
  - $h_{\frac{1}{2}} = \frac{\log(0.5)}{\log(\alpha+\beta)}$
  - Example: SP500
    - $\alpha + \beta \approx 0.97 - 0.99$
    - Half-lives of 23-69 trading days (1 to 3 months)

The GARCH(1,1) model

- How we get GARCH(1,1) coefficients
  - Maximum likelihood (ML) is not possible: σ_t^2 enters the likelihood calculation — circularity problem
    - $\mathcal{L}_{t(\theta)} = -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log\left(\sigma_t^2(\theta)\right) - \frac{\varepsilon_t^2}{2\sigma_t^2(\theta)}$, θ: all coefficients (ω, α, β)
  - Use **Quasi-maximum likelihood** (QML)! ☺
    - QML treats the Gaussian likelihood as a working approximation — the variance parameters are consistently estimated even when returns are fat-tailed, as long as we use robust (sandwich) standard errors
    - In practice, use an estimator of the standard errors

Figure 9.4 — GARCH(1,1) conditional standard deviation, 1980–2019



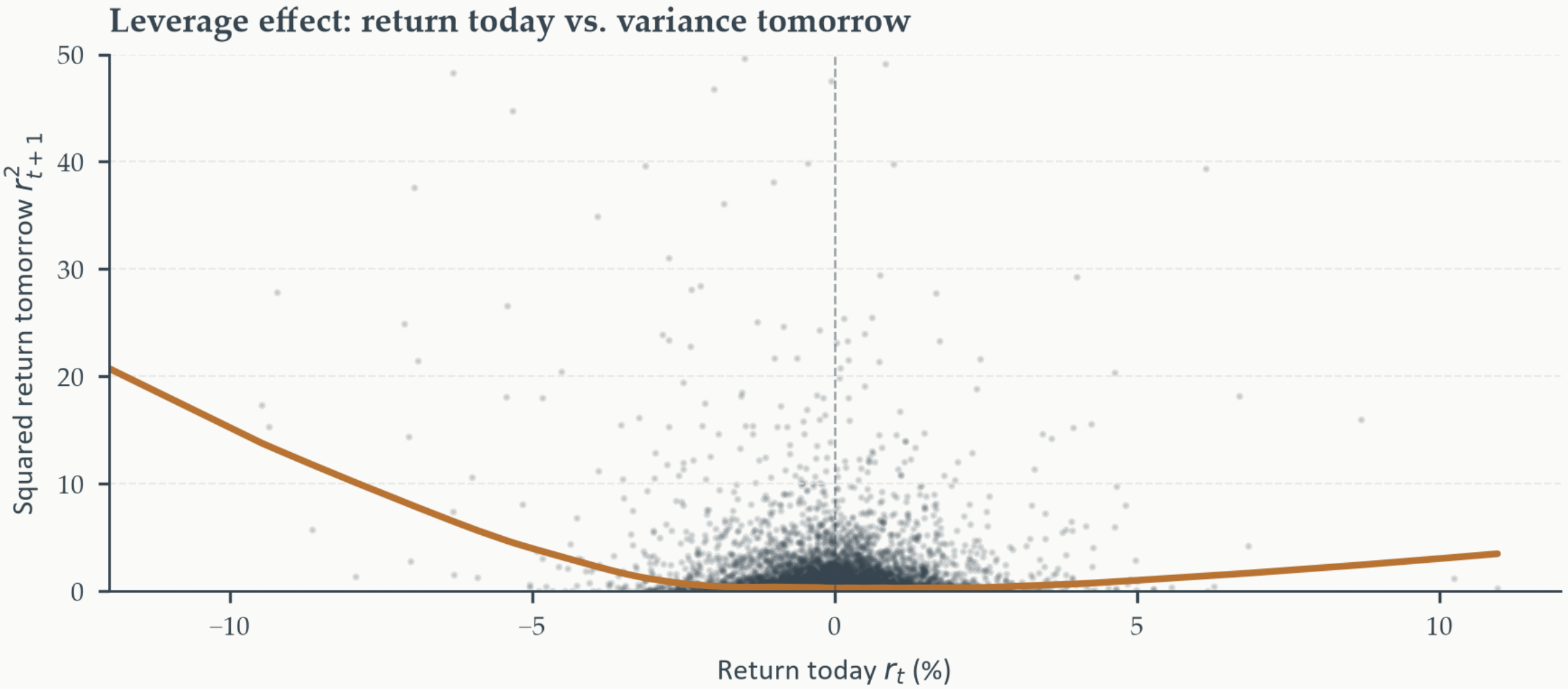
Asymmetric volatility

Section 4

Asymmetric volatility

- GARCH(1,1) assumes that a -3% and a 3% shock have the same impact on volatility
  - The leverage effect: Negative shocks should raise future volatility more than positive shocks
    - A price decline raises the firm's debt-to-equity ratio, making it riskier
    - A failing price raises the risk premium investors demand, and this raises volatility
  - Is the leverage effect observed in the data

Figure 9.6 — The leverage effect in S&P 500 returns



Asymmetric volatility

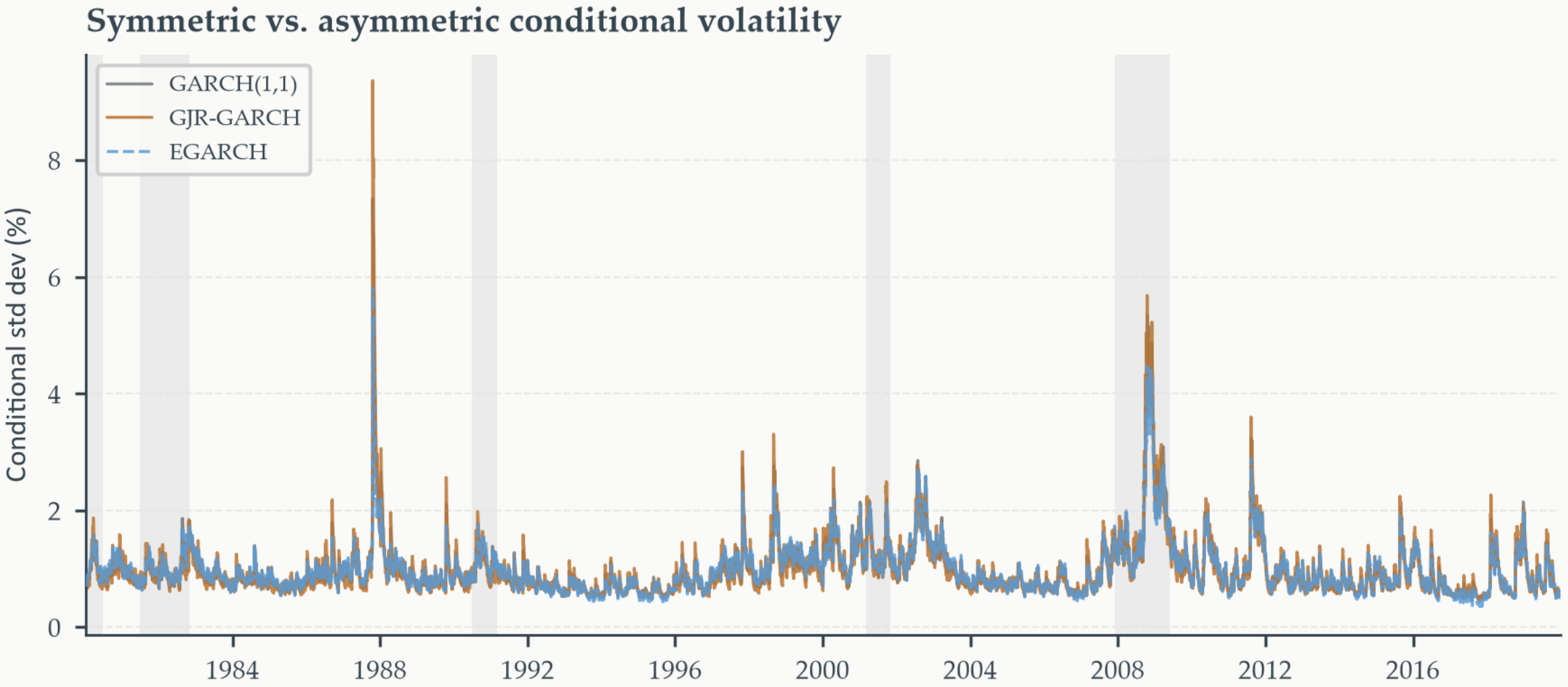
- GJR-GARCH model
  - Add a term that switches on for negative shocks (bad news)
  - $\sigma_t^2 = \omega + \left(\alpha + \gamma D_{\varepsilon_{t-1}}\right)\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$

    - $D = \begin{cases} 1 & if\ \varepsilon_{t-1} < 0 \\ 0 & if\ \varepsilon_{t-1} \geq 0 \end{cases}$

Asymmetric volatility

- EGARCH model
  - The GJR-GARCH model requires $\omega, \alpha, \beta, \gamma > 0$
  - The EGARCH model is alternative that faces less coefficient constraints
  - Use the log of the **conditional** variance
    - $\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \alpha |z_{t-1}| + \gamma z_{t-1}, \quad z_{t-1} = \frac{\varepsilon_{t-1}}{\sigma_{t-1}}$
  - $\sigma_t^2 > 0$ is guaranteed by construction — because we model $\log \sigma_t^2$
    - The coefficients themselves don't face a sign restriction

Figure 9.7 — GARCH(1,1) vs. GJR-GARCH vs. EGARCH conditional volatility



Dynamic conditional correlation

Section 5

Dynamic conditional correlation

- If during a crisis two assets become more volatile, does that make the portfolio more or less risky?
- The answer depends on what happens to the correlation between both assets
  - During a crisis, correlation between risky assets tends to increase — precisely when diversification is needed the most
  - Models assuming a constant correlation miss this dynamic, leading to under-hedging during episode when this is most important

Dynamic conditional correlation

- The Two-Step DCC procedure
  - We need a matrix that captures correlation changes across different assets (H_t)
  - Let $H_t = D_t R_t D_t$
    - D_t: diagonal (conditional) standard deviations
    - R_t: the conditional correlation matrix

Dynamic conditional correlation

- The Two-Step DCC procedure (cont.)
  - You have n assets
  - Step 1: Fit a separate GARCH(1,1) to each series and extract standardized residuals: $\hat{z}_{it} = \frac{\hat{\varepsilon}_{it}}{\hat{\sigma}_{it}}$
  - Step 2: Build a matrix D_t
    - $D_t = \begin{bmatrix} \hat{z}_{1t} & 0 & 0 & 0 \\ 0 & \hat{z}_{2t} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \hat{z}_{nt} \end{bmatrix}$
  - Step 3: Model the correlation among the $\hat{z}_{it}$ via a scalar recursion

Dynamic conditional correlation

- The Two-Step DCC procedure (cont…)
  - The correlation recursion
    - $Q_t = (1 - a - b)\bar{Q} + a\boldsymbol{z}_{t-1}\boldsymbol{z}'_{t-1} + bQ_{t-1}$
    - a: how fast Q_t responds to a new shock; b: how much of Q_{t-1} carries forward
    - Q_t is an auxiliary positive-definite matrix — it is **not** a correlation matrix yet (its diagonal entries are not 1)
    - $\bar{Q}$ is estimated from the data as the sample covariance of the standardized residuals $\hat{z}_{It}$ — the long-run average correlation
    - a and b are estimated by QML in Step 2, the same way α and β are estimated in GARCH(1,1)

Dynamic conditional correlation

- The Two-Step DCC procedure (cont…)
  - Q_t has the right covariance structure but the wrong scale — its diagonal elements are not 1
  - Dividing by $\sqrt{q_{ii,t}} \cdot \sqrt{q_{jj,t}}$ forces the diagonal to 1 and maps all off-diagonal entries (-1,1)
  - *The result* R_t is a proper correlation matrix at every t
  - $R_t = \mathrm{diag}(Q_t)^{-\frac{1}{2}} Q_t \, \mathrm{diag}(Q_t)^{-\frac{1}{2}}$
  - Step 4: Obtain H_t
    - $H_t = D_t R_t D_t$

Dynamic conditional correlation

- The Two-Step DCC procedure (cont…)
  - Example
    - $\bar{Q} = Q_{t-1} = \begin{bmatrix} 1.00 & 0.20 \\ 0.20 & 1.00 \end{bmatrix}, a = 0.05, b = 0.93$
    - Size of shocks: $z_{1,t-1} = -2.0, z_{2,t-1} = -2.5$
    - $\boldsymbol{z}_{t-1}\boldsymbol{z}'_{t-1} = \begin{bmatrix} 4.00 & 5.00 \\ 5.00 & 6.25 \end{bmatrix}$
    - The off diagonal elements drives the correlation update between both stocks
    - $Q_t = (1 - a - b)\bar{Q} + a\boldsymbol{z}_{t-1}\boldsymbol{z}'_{t-1} + bQ_{t-1}$
    - $Q_t = 0.02\begin{bmatrix} 1.00 & 0.20 \\ 0.20 & 1.00 \end{bmatrix} + 0.05\begin{bmatrix} 4.00 & 5.00 \\ 5.00 & 6.25 \end{bmatrix} + 0.93\begin{bmatrix} 1.00 & 0.20 \\ 0.20 & 1.00 \end{bmatrix} = \begin{bmatrix} 1.15 & 0.44 \\ 0.44 & 1.26 \end{bmatrix}$
    - $R_t = \mathrm{diag}(Q_t)^{-\frac{1}{2}} Q_t \,\mathrm{diag}(Q_t)^{-\frac{1}{2}} = \begin{bmatrix} 1.000 & 0.365 \\ 0.365 & 1.000 \end{bmatrix}$
    - Conditional correlation jumps from 0.20 to 0.365

GARCH-based Value-at-Risk

Section 6

GARCH-based Value-at-Risk

- What's Value-at-Risk (VaR)?
  - Standard risk measure: The standard deviation
    - The expected deviation from the mean
  - VaR: Given a confidence level, how much value is at risk?
    - VaR: The loss not exceeded with probability $1 - \alpha$ over horizon h
    - VaR(1%) is the loss level exceeded on only 1% of trading days

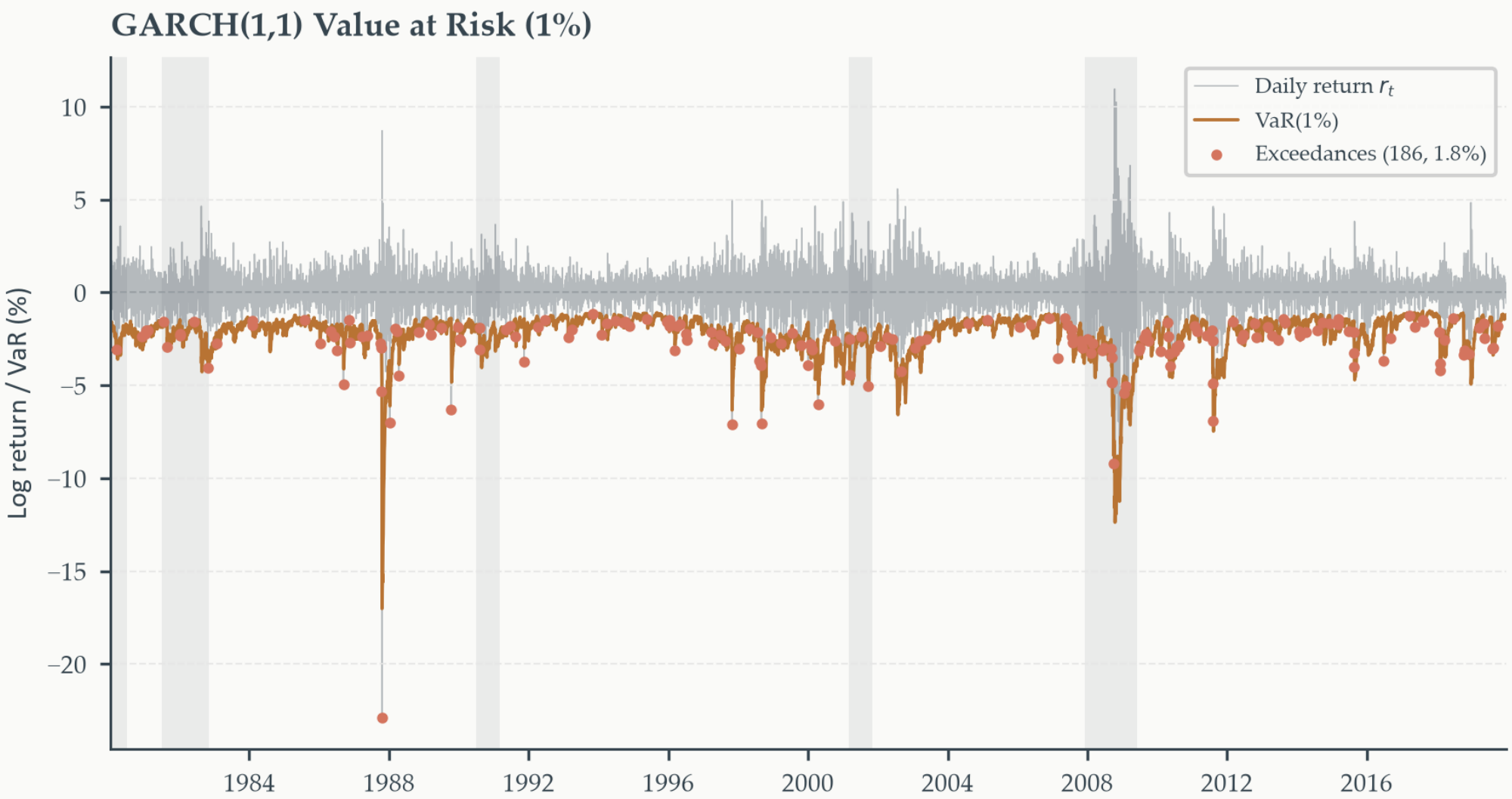
GARCH-based Value-at-Risk

- VaR vs historical data
  - Historical data: Sort the past 250 daily returns, cut off at your confidence level
    - This approach assigns the same weight to all observations — a calm Tuesday 5 years ago has the same relevance as yesterday
    - During calm periods: Risk is overstated (old crises remain in the sample)
    - During a crisis: Risk is understated (calm periods remain on the sample)

GARCH-based Value-at-Risk

- Parametric GARCH-VaR
  - Create a dynamic/updating VaR
  - Remember: $r_{t+1} = \mu + \sigma_{t+1} z_{t_1}$ with $z_{t+1} \sim N(0,1)$
  - $VaR_{t+1}(1\%) = \hat{\mu} - 2.326\hat{\sigma}_{t+1}$
  - Your VaR estimate updates daily
    - It contracts during calm periods
    - It expands during crises

Figure 9.8 — GARCH(1,1) one-day 1% Value at Risk, 1980–2019



Volatility Models

Chapter 9

Nicolas Cachanosky

Department of Economics and Finance